

Business Dataset Preprocessing — Analysis Report

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Dataset Overview

File: Business_dataset.csv

Rows: 4,362

Columns: 5 — Category, Price, Rating, Stock, Discount

Data types: 2 categorical (Category, Stock), 3 numeric (Price, Rating, Discount)

Task 1: Missing Value Treatment

Missing Value Analysis

Category 2,830 64.9% Mode imputation Price 64 1.5% Median imputation Rating 1,039 23.8% Mean imputation Stock 1,807 41.4% Mode imputation Discount 622 14.3% Median imputation

Methodology and Justification

1. Category — Mode Imputation

Method: Filled missing values with the most frequent category.

Reason: As a nominal categorical variable, no arithmetic operation (mean/median) is valid. Mode preserves the existing category distribution without introducing artificial categories.

Data quality impact: Maintains class distribution integrity; avoids misclassification bias.

2. Price — Median Imputation

Method: Filled missing values with the median price.

Reason: Price contains a wide range (approx. 100–10,000) and likely has outliers. Median is robust to extreme values, unlike the mean which would be pulled by high-end prices.

Data quality impact: Preserves central tendency without distortion from outliers.

3. Rating — Mean Imputation

Method: Filled missing values with the mean rating.

Reason: Rating values are bounded (~1–5) and appear roughly symmetrically distributed. Mean is the most efficient estimator for symmetric distributions with no severe outliers.

Data quality impact: Provides the best unbiased estimate of the typical rating.

4. Stock — Mode Imputation

Method: Filled missing values with the most frequent stock status.

Reason: Binary categorical variable where mode simply assigns the majority class, which is the safest single guess for a categorical variable.

Data quality impact: Preserves the majority-class proportion while filling unknowns.

5. Discount — Median Imputation

Method: Filled missing values with the median discount.

Reason: Discount values range from 0–49 with many zeros (no discount) and a right-skewed distribution. Median avoids the inflation that mean would cause from high discounts.

Data quality impact: More representative of the typical discount value.

Task 2: Data Normalization

Technique Applied

Min-Max Normalization (also called Min-Max Scaling)

Formula:

$$X_{\text{scaled}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

Columns Normalized

Price ~100 – 10,000 [0, 1] Rating ~1.0 – 5.0 [0, 1] Discount 0 – 49 [0, 1]

Why Min-Max Normalization?

- **Comparable scale:** All three numeric features are scaled to the $[0,1]$ range, making their magnitudes directly comparable. Without normalization, Price (thousands) would dominate Rating ($\sim 1-5$) in any distance-based or gradient-based algorithm.
- **Preserves distribution shape:** Unlike standardization (z-score), Min-Max preserves the exact relative distances between data points. A product with twice the price of another will still have twice the scaled value.
- **Bounded output:** Values are confined to $[0,1]$, which is ideal for models that expect inputs in this range (neural networks, regularization-based models).
- **Interpretability:** The scaled value directly represents where a sample falls within the observed range — e.g., a Price of 0.5 means it is halfway between the cheapest and most expensive product.
- **No distributional assumptions:** Unlike z-score normalization (which assumes approximately normal data), Min-Max works regardless of the underlying distribution.

Why Not Z-Score Standardization?

Z-score would center data at 0 with unit variance, but:

- It does not bound values — outliers can produce extreme z-scores.
- Negative values are less intuitive to interpret.
- It assumes at least roughly normal data, which Discount (with many zeros) violates.

Why Not Robust Scaling?

Robust scaling (using median and IQR) is useful when outliers are extreme, but the ranges in this dataset are well-behaved and bounded, so Min-Max is sufficient and more interpretable.

Summary

Missing values (categorical) Mode imputation Preserves distribution Missing values (numeric) Median imputation Robust to outliers Missing values (rating) Mean imputation Efficient for symmetric data Normalization Min-Max scaling Unifies scale to $[0,1]$

All preprocessing decisions were made to maximize data quality, preserve statistical properties, and ensure downstream model compatibility.